```java
public class Main{
    public static void main(String[] aStrings){
        // #1
        String name = "SEAN Manutnithya";
        System.out.println("Hello " + name);
        displayP(aStrings);
        displayShapeA(aStrings);
        displayShapeB(aStrings);
        displayC(aStrings);
        tableA(aStrings);
        tableB(aStrings);
        System.out.print("Text length is: " + textLen(aStrings));
        System.out.print("Result: " + cutText(aStrings));
        findVowelIdx(aStrings);
        textCompare(aStrings);
        textContain(aStrings);
        Student std1 = new Student(0002, "Makara", 'F', 17, 88.3, 4);
        Student std2 = new Student(0001, "Tola", 'M', 15, 98.3, 3);
        std1.display();
        std2.display();
    }
 output: Hello SEAN Manutnithya
    // #2
    public  static void displayP(String[] aStrings){
        System.out.println("\n Line break");
        System.out.println("\t Tabulation");
        System.out.println("\' Single Quote");
        System.out.println("\" Double Quote");
        System.out.println("\\  \\Sign");
        System.out.println("\\\\ \\\\Sign");
        System.out.println("// Line command");
        System.out.println("/*...*/ Block command");
        System.out.println("\"\"\" \n \"\"\" text block");

    }
 output:

 Line break
       Tabulation
' Single Quote
" Double Quote
\  \Sign
\\ \\Sign
// Line command
```

```
/*...*/ Block command
"""
 """ text block

    // #3
    public static  void displayShapeA(String[] asStrings){
        String str = "*";
        int extraSt = 0;
        for(int i = 21; i > 0;i-=2){
            String repeat = str.repeat(i);
            System.out.println(" ".repeat(extraSt++) + repeat);
        }
    }
    public  static void displayShapeB(String[] aStrings){
        int len = 17;
        int height = 10;
        String str = "*";
        String strLen = "\n*\t\t*";
        System.out.print(str.repeat(len));
        System.out.print(strLen.repeat(height));
        System.out.print("\n" + str.repeat(len));
    }
    public static void displayC(String[] aStrings){
        for(int r = 0; r < 5; r++){
            for(int c = 0; c < 5 ; c++){
                System.out.print(r + c + 1 + " ");
            }
            System.out.println();
        }
    }
 output:
**********
 **********
  **********
   **********
    **********
     **********
      *********
       *******
        *****
         ***
          *
**********
*          *
*          *
```

```
*           *
*           *
*           *
*           *
*           *
*           *
*           *
*           *
**************
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
```

```java
    // #4
    public static void tableA(String[] aStrings){
        System.out.print("\t |");
        for(int row=1; row<=10;row++){
            System.out.print("\t" + row + "|");
        }
        System.out.print("\n"+"-".repeat(90) + "\n");
        // System.out.println();
        int len = 0;
        for(int row = 1; row <= 5; row++){
            for(int col = 1; col <= 10; col++){
                int num = row * col;
                if(num < 10 && num % 10 != 0){
                    len = 1;
                }
                if(col == 1) System.out.print("\t" + row + "|");
                if(len == 1){
                    System.out.print("\t" + " " + num + "|");
                    len = 0;
                }else {
                    System.out.print("\t" + num + "|");
                }
            }
            System.out.println();
        }
    }
    public static void tableB(String[] aStrings){
        System.out.print("""
            +----------------+-----------+--------------------+---------+
            | Country        | Name      | Professoin         | Age     |
```

```
            +-----------------+-----------+---------------------+---------+
            | Germany         | Micheal   | Computer Engineer   | 19      |
            +-----------------+-----------+---------------------+---------+
            | England         | Robert    | Artist              | 34      |
            +-----------------+-----------+---------------------+---------+
            | United Kingdom  | Julia     | Designer            | 42      |
            +-----------------+-----------+---------------------+---------+
            | United State    | Jo        | Actor               | 21      |
            +-----------------+-----------+---------------------+---------+
        """ );

    }
output:
         |     1|     2|     3|     4|     5|     6|     7|     8|     9|     10|
  ----------------------------------------------------------------------------------
        1|     1|     2|     3|     4|     5|     6|     7|     8|     9|     10|
        2|     2|     4|     6|     8|    10|    12|    14|    16|    18|     20|
        3|     3|     6|     9|    12|    15|    18|    21|    24|    27|     30|
        4|     4|     8|    12|    16|    20|    24|    28|    32|    36|     40|
        5|     5|    10|    15|    20|    25|    30|    35|    40|    45|     50|
      +-----------------+-----------+---------------------+---------+
      | Country         | Name      | Professoin          | Age     |
      +-----------------+-----------+---------------------+---------+
      | Germany         | Micheal   | Computer Engineer   | 19      |
      +-----------------+-----------+---------------------+---------+
      | England         | Robert    | Artist              | 34      |
      +-----------------+-----------+---------------------+---------+
      | United Kingdom  | Julia     | Designer            | 42      |
      +-----------------+-----------+---------------------+---------+
      | United State    | Jo        | Actor               | 21      |
      +-----------------+-----------+---------------------+---------+
    // #5
    public static String textLen(String[] aStrings){
        String text = "I love my Home Country";

        int len =  text.length();
        return String.valueOf(len);
    }
    public static String cutText(String[] aStrings){
        String text = "I love my little country";
        String cutText = text.substring(7);

        return cutText;
    }
    public static void findVowelIdx(String[] aStrings){
        String text = "Hi Student!";
        String vols = "aeiouAEIOU";

        for(int t=0; t<text.length(); t++){
            char ch = text.charAt(t);
            if(vols.indexOf(ch) != -1){
```

```
                System.out.println("\"" + ch + "\" is at index: " + t);
            }
        }
    }
    public static void textCompare(String[] aStrings){
        String text1 = "Hi Students!";
        String text2 = "Students!";
        String text3 = "Hi Students!";
        System.out.println("text1 is eqaul to text2: " + text1.equals(text2));
        System.out.println("text1 is eqaul to text3: " + text1.equals(text3));
    }
    public static void textContain(String[] aStrings){
        String text1 = "Hi Students!";
        String text2 = "Students";
        String text3 = "Teacher";
        System.out.println("text1 contains text2: " + text1.contains(text2));
        System.out.println("text1 contains text3: " + text1.contains(text3));
    }
}
output:
    Text length is: 22Result: my little country"i" is at index: 1
    "u" is at index: 5
    "e" is at index: 7
    text1 is eqaul to text2: false
    text1 is eqaul to text3: true
    text1 contains text2: true
    text1 contains text3: false
    // #6
    class Car{
        String model;
        String color;
        String year;
        int price;
        int warenty;
    }
    class Employee {
        String name;
        String Id;
        String DOB;
        String dept;
        String offi;
    }
    class BankAccount {
        String name;
        String password;
```

```
        String accNum;
        int balance;
        int expense;
    }
    class Video{
        String name;
        String ext;
        int length;
        int resolution;
        int size;
    }
    class Student{
        int studentId;
        String name;
        char gender;
        int age;
        double score;
        int year;
        Student(int studentId, String name, char gender, int age, double score,
int year){
            this.studentId = studentId;
            this.name = name;
            this.gender = gender;
            this.age = age;
            this.score = score;
            this.year = year;
        }
        void display(){
            System.out.printf(
            """
                StudentId: %d
                Name:      %s
                Gender:    %c
                Age:       %d
                Score:     %.1f
                Year:      %d
            """,studentId,name,gender,age,score,year);
        }
    }
output:
```

```
StudentId: 2
Name:      Makara
Gender:    F
Age:       17
Score:     88.3
Year:      4
StudentId: 1
Name:      Tola
Gender:    M
Age:       15
Score:     98.3
Year:      3
```